

KARAN RAJENDRA

SOFTWARE ENGINEER

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EDUCATION

Stony Brook University

Master of Science in Computer Engineering — GPA: 3.9/4.0

May 2026

Stony Brook, NY

Coursework: Distributed Systems, Advanced Algorithms, Machine Learning, Deep Learning, Natural Language Processing

Visvesvaraya Technological University

Bachelor of Engineering in Computer Science

Aug 2019 – Aug 2023

Bangalore, India

WORK EXPERIENCE

Software Engineer

Aug 2023 – June 2024

Meta

Bangalore, India

- Engineered an end-to-end data pipeline and clustering framework for Instagram, processing **1.2B** user pairs to identify **500M** unique users and enabling the platform's first cohort analysis.
- Optimized large-scale data processing workflows by re-engineering pipeline logic, improving performance by **5x** while reducing Business Compute Unit (BCU) consumption by **50%**.
- Developed Instagram-specific behavioral signals, including story interactions, and implemented multi-signal clustering logic to address data sparsity and improve user segmentation.
- Designed and built scalable monitoring dashboards to track feature adoption and experiment performance, enabling data-driven product decisions across cross-functional teams.

Research Assistant

Jan 2025 – Present

Stony Brook University

Stony Brook, NY

- Built distributed backend services in **C++** and **Python**, improving throughput by **25%** using multithreading, asynchronous processing, and scalable scheduling.
- Developed fault-tolerant microservices using **Kafka**, **gRPC**, and **Docker** with replication and automatic failover mechanisms.
- Designed and optimized backend APIs and distributed communication pipelines for reliable large-scale system coordination.
- Profiled and debugged Linux applications using **perf**, **Valgrind**, and **GDB**, reducing runtime overhead by **20%**.
- Built CI/CD pipelines, monitoring, and observability workflows for reproducible multi-node deployments and faster debugging.
- Collaborated with researchers and engineers to improve distributed infrastructure reliability, scalability, and operational efficiency.

Graduate Teaching Assistant

Aug 2024 – Dec 2024

Stony Brook University

Stony Brook, NY

- Taught core distributed systems concepts including consensus, replication, sharding, fault tolerance, and scalability using Python and C++ based implementations.
- Designed and graded assignments focused on distributed system trade-offs, reliability, debugging, and performance optimization, improving project quality by **25%**.

PROJECTS

AWS Cloud Infrastructure Kubernetes Deployment

AWS, Kubernetes, Terraform, GitHub Actions, CI/CD [\[GitHub\]](#)

- Designed and deployed a highly available cloud infrastructure on AWS using Amazon EKS across multiple Availability Zones.
- Automated infrastructure provisioning and deployment using **Terraform**, **GitHub Actions**, and **Docker**.
- Configured Route53, Application Load Balancers, Auto Scaling, CloudFront, and RDS to support scalable, fault-tolerant services.
- Implemented end-to-end CI/CD pipelines, reducing deployment effort and enabling reliable automated releases.

Speech Emotion Detection Application

Deep Learning, FastAPI, Real-Time Inference, Google Cloud Run [\[Demo\]](#)

- Developed a speech emotion classifier using a **CNN-BiLSTM-Attention** model deployed on GCP via Docker.
- Built a FastAPI backend and React frontend, styled with TailwindCSS and Framer Motion for an engaging user experience.
- Reduced API response latency by **40%** (**650ms** → **390ms**) through asynchronous I/O, serving **100+** real-time inference requests.

SKILLS

Languages: Python, C++, Java, SQL, JavaScript, TypeScript, Scala

Backend Distributed Systems: FastAPI, Flask, Node.js, REST APIs, Kafka, gRPC, Microservices

Cloud DevOps: AWS, GCP, Azure, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, Airflow

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face, NLP, Deep Learning, Transformers, LoRA

Data Engineering Big Data: Spark, Hive, HDFS, PostgreSQL, MySQL, MongoDB, Redis

Monitoring Performance: Linux, GDB, Valgrind, perf, TensorBoard, Weights Biases

Visualization: Tableau, Matplotlib, Power BI